

REVISION - SOFTWARE QUALITY ASSURANCE

Question 1: Software Testing Strategy

The testing team experienced inconsistent testing practices throughout the Bank Management System project.

1. Explain four reasons why every software project should have a well-defined **Software Testing Strategy**.
2. Recommend three testing strategies, testing concepts, or testing techniques that should be adopted for the Bank Management System. Justify each recommendation.

Question 2: Project Documentation and Quality Assurance

Effective project documentation is essential for successful software development and quality assurance.

1. Identify and explain five project documents that should have been prepared and maintained during the development of the Bank Management System.
2. Discuss how each of these documents contributes to improving software quality, communication, and project success.

Question 3: Contract Review

Using the case study, answer the following questions.

1. Identify and explain four project problems that could have been prevented through an effective **Contract Review** process.
2. Explain three objectives of **Contract Review** and discuss how each objective would contribute to the successful delivery of the Bank Management System.

Question 4: Software Testing Implementation

As the **Lead Software Test Engineer**, design a structured **Software Testing Implementation Process** for the Bank Management System.

Your answer should include:

- Test Planning
- Test Environment Setup
- Test Data Preparation
- Test Case Development
- Test Execution
- Defect Reporting and Tracking
- Retesting and Regression Testing
- Test Reporting and Documentation
- Test Closure

Question 5: Software Quality Metrics and Testing Evaluation

Before approving the Bank Management System for deployment, management requires objective evidence of software quality.

1. Identify and explain five software quality metrics or **Key Performance Indicators (KPIs)** that should be used to evaluate the testing process.
2. Explain how these metrics assist project managers in making informed software release decisions.

Question 6: Development and Quality Planning

The project was initiated without adequate planning documents.

1. Explain the importance of preparing both a **Software Development Plan (SDP)** and a **Software Quality Assurance Plan (SQAP)** before software development begins.
2. Compare the **Software Development Plan (SDP)** and the **Software Quality Assurance Plan (SQAP)** by discussing six major differences between the two documents.

Question 7: Test Environment and Test Data Management

The case study indicates that the testing environment differed from the production environment and that test data were inadequate.

1. Explain five characteristics of an effective **Software Test Environment**.
2. Discuss five best practices for preparing and managing **Test Data** to ensure effective software testing.

Question 8: Defect Management and Risk Management

1. Identify and explain the major risks that may arise if the Bank Management System is deployed without adequate **Software Quality Assurance** activities.
2. Explain how an effective **Defect Management Process** improves software quality by discussing the following stages:
 - Defect Identification
 - Defect Reporting
 - Defect Tracking
 - Defect Resolution
 - Defect Verification and Closure

Question 9: Test Reporting and Software Release Decision

The project manager has requested a recommendation on whether the Bank Management System is ready for deployment.

1. Describe the information that should be included in a comprehensive **Test Summary Report** before approving software deployment.
2. Discuss the factors that should be considered before making the final **Software Release Decision**.

Question 10: Software Quality Improvement Plan

Prepare a comprehensive **Software Quality Improvement Plan** by recommending appropriate actions in the following areas:

- Contract Review
- Software Development Planning
- Software Quality Assurance Planning
- Software Testing Strategy
- Software Testing Implementation
- Defect Management
- Test Reporting and Documentation
- Continuous Quality Improvement

END.